

Michael Fehl

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EDUCATION

Barcelona School of Economics

Barcelona, Spain

Master Program in Economics & Finance

09/2025 – 07/2026

Relevant Courses: Advanced Panel Data Methods, Nonparametric Methods & Distributional Analysis, Now-casting & Forecasting, Network Econometrics, Network Algorithms, Geospatial DS & Economic Spatial Models

Thesis: Who Absorbs the Remittance Shock: The Migrant Worker or the Household?

Degree awarded by Universitat Pompeu Fabra and Universitat Autònoma de Barcelona.

University of North Carolina at Chapel Hill

Chapel Hill, USA

Bachelor of Science in Economics

08/2019 – 05/2023

Philosophy, Politics, and Economics Minor

EXPERIENCE

Measuring Migrant Remittance Flows using DiD and Network Modelling

Barcelona, ES

Master's Thesis | Python, SQL, Git

02/2026 – Present

- Engineered a SQL data pipeline consolidating 700+ Excel tables of Mexican municipal migration flows across 50 US states and 14 years into DuckDB for panel analysis
- Constructed weighted allocation matrices to impute bilateral remittance flows across 2,500+ municipalities
- Modeled Mexico-US migration as a weighted bipartite network; traced how localized US policy shocks propagate to non-exposed Mexican municipalities through intermediary nodes
- Applied difference-in-differences to estimate policy shock transmit to Mexican household remittance income

Resource Allocation under Uncertainty & Network Dynamics

Personal Project

Independent Researcher | MATLAB

12/2025 – 02/2026

- Developed a general equilibrium model in MATLAB to simulate multi-agent trade under uncertainty.
- Programmed algorithms to solve for Walrasian equilibria and the core, validated against welfare theorems
- Ran simulations on different starting prices to visualize tatonnement process as an invisible auctioneer
- Constructed 3D spatiotemporal visualizations to track the density of goods & utility across different starting prices and allocations

Duke University

Durham, USA

Research Assistant | Excel, R

06/2024 – 01/2025

- Imported & cleaned raw economic data from BEA and FRED using R to model industry profit trends.
- Investigated key macro indicator trends during recessions & expansions in Excel & R.
- Designed interactive dashboards to display updated macro indicators and corporate profits.
- Developed automated tables in Excel that refresh macro data and dynamically update charts.

University of North Carolina at Chapel Hill

Chapel Hill, USA

Independent Researcher | R

02/2024 – 04/2024

- Cleaned & analyzed a dataset of 32,000,000 observations in R using efficient data-handling.
- Deployed machine learning models such as Random Forests, SVM, and boosting to optimize predictions.
- Implemented an XGBoost classifier that achieved 88.4% accuracy in predicting customer behavior.
- Visualized model interpretability, hyperparameter tuning, and feature importance.

SKILLS & LANGUAGES

Analytical Methods Causal Inference, Time Series Forecasting, Econometrics, Machine Learning

Software & Tools R, Python, Stata, MATLAB, SQL, Excel, L^AT_EX, Quarto, Git

Languages English (Native), Spanish (Intermediate)

ADDITIONAL INFORMATION

- Personal Website** Barcelona, Spain (2025)
Built a personal website from scratch using Quarto, Git, and SCSS to host my portfolio.